applied sciences

MDPI

Review

Vocal Communication Between Cobots and Humans to Enhance Productivity and Safety: Review and Discussion

Yuval Cohen [1,*], Maurizio Faccio [2] and Shai Rozenes [3]

[1] Industrial Engineering Department, Tel Aviv Afeka Academic College of Engineering, Tel Aviv 69988, Israel
[2] Department of Technical Management of Industrial Systems, University of Padova, 35122 Padova, Italy; maurizio.faccio@unipd.it
[3] Industrial Engineering and Technology Management Department, Holon Institute of Technology, Holon 58102, Israel; rozeness@hit.ac.il
* Correspondence: yuvalc@afeka.ac.il; Tel.: +972-545398152

Abstract: This paper explores strategies for fostering efficient vocal communication and collaboration between human workers and collaborative robots (cobots) in assembly processes. Vocal communication enables the division of attention of the worker, as it frees their visual attention and the worker's hands, dedicated to the task at hand. Speech generation and speech recognition are pre-requisites for effective vocal communication. This study focuses on cobot assistive tasks, where the human is in charge of the work and performs the main tasks while the cobot assists the worker in various peripheral jobs, such as bringing tools, parts, or materials, and returning them or disposing of them, or screwing or packaging the products. A nuanced understanding is necessary for optimizing human–robot interactions and enhancing overall productivity and safety. Through a comprehensive review of the relevant literature and an illustrative example with worked scenarios, this manuscript identifies key factors influencing successful vocal communication and proposes practical strategies for implementation.

Keywords: cobot; human–robot communication; vocal communication; vocal interface; auditory communication; auditory interface; collaborative robotics; Industry 5.0

check for updates

Academic Editor: Douglas O'Shaughnessy

Received: 18 November 2024
Revised: 24 December 2024
Accepted: 6 January 2025
Published: 13 January 2025

Citation: Cohen, Y.; Faccio, M.; Rozenes, S. Vocal Communication Between Cobots and Humans to Enhance Productivity and Safety: Review and Discussion. *Appl. Sci.* **2025**, *15*, 726. https://doi.org/10.3390/app15020726

Copyright: © 2025 by the authors. Licensee MDPI, Basel, Switzerland. This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (https://creativecommons.org/licenses/by/4.0/).

1. Introduction

Cobots operate in shared workspaces with humans, necessitating seamless interaction with their human counterparts [1]. This interaction goes beyond mere task execution; it involves a dynamic exchange of information, coordination of efforts, and mutual understanding to optimize the synergies between human skills and robotic precision [2].

Effective communication and collaboration between human workers and cobots are pivotal for maximizing the benefits of cobot integration. The need arises from the intricacies of shared tasks, where human operators and cobots must coordinate their actions to achieve optimal results [3]. Moreover, as cobots are designed to be more adaptable and responsive to changes in the manufacturing environment, clear communication becomes essential for successful task allocation, problem-solving, and overall operational efficiency [4]. Effective communication fosters a sense of cooperation, trust, and shared purpose, thereby enhancing the overall productivity and safety of the collaborative work environment [5].

This paper focuses on human–cobot communication related to cobots' assistive functions. Vocal communication can free a worker's hands and eyes, allowing them to perform tasks while receiving information or instructions through audio support [6]. This approach is preferred over augmented reality (AR) or head-mounted display-based support in many

application areas today [7]. So, for assisting a worker performing manufacturing or assembly tasks, vocal and audio communication are preferred over other communication forms.

This paper delves into strategies for optimizing vocal communication and collaboration between human workers and cobots. It shows that these strategies enhance productivity and safety in assistive work. This is its contribution to the transformative role of collaborative robotics in modern manufacturing. By addressing this critical nexus, we aim to guide future research endeavors and industry practices toward a better collaboration of humans and cobots on the factory floor.

This paper continues as follows: Section 2 presents a literature review. Section 3 defines the main cobot assistive tasks in industrial settings. Section 4 defines proposed strategies for optimizing vocal communication, Section 5 presents a case study, Section 6 discusses an illustrative example, and Section 7 concludes the paper.

2. Literature Review

Human–cobot communication plays an important role in task allocation and the safety of collaborative assembly systems [8,9]. Schmidbauer et al. [10] also examined the static vs. dynamic task allocation preferences of human workers and found that they preferred adaptive task sharing (ATS) to static predetermined task allocation and reported increased satisfaction with dynamic allocation. Workers are more likely to assign manual tasks to cobots, while preferring to handle cognitive tasks themselves [10]. These dynamic characteristics of collaboration require good communication between humans and cobots. Cobot–human communications could take one of the following forms (as summarized in Table 1):

- *Text communications*: The human worker uses text for programming the cobot, and to send it commands and information [11]. The cobot may also use text to communicate messages, explanations, cautionary warnings, and other information types [12,13].
- *Visual communications*: Graphical displays, AR, and vision systems are key components of visual interfaces that enable workers to comprehend and interpret information from cobots efficiently [14,15]. Graphical displays can provide real-time feedback on cobot actions, task progress, and system status, enhancing transparency and situational awareness [16]. AR overlays digital information onto the physical workspace, offering intuitive guidance for tasks and aiding in error prevention [17]. Vision systems, equipped with cameras and sensors, enable cobots to recognize and respond to human gestures, further fostering natural and fluid interaction [18].
- *Auditory communications*: Human–cobot vocal communication has been a topic of intensive research in recent years [19]. Auditory cues are valuable in environments where visual attention may be divided or compromised [20]. Sound alerts, spoken instructions, and auditory feedback mechanisms contribute to effective communication between human workers and cobots [21]. For instance, audible signals can indicate the initiation or completion of a task, providing workers with real-time information without requiring constant visual focus [22]. Speech recognition technology enables cobots to understand verbal commands, fostering a more intuitive and dynamic interaction [23]. The thoughtful use of auditory interfaces between humans and cobots helps create a collaborative environment where information is conveyed promptly, enhancing overall responsiveness and coordination [24]. Several recent papers have proposed novel interfaces and platforms to facilitate this type of interaction. Rusan and Mocanu [25] introduced a framework that detects and recognizes speech messages, converting them into spoken commands for operating system instructions. Carr, Wang, and Wang [26] proposed a network-independent verbal communication platform for multi-robot systems, which can function in environments lacking network infrastructures. McMillan et al. [27] highlighted the importance of conversation as a

natural method of communication between humans and robots, promoting inclusivity in human–robot interaction. Lee et al. [28] conducted a user study to understand the impact of robot attributes on team dynamics and collaboration performance, finding that vocalizing robot intentions can decrease team performance and perceived safety. Ionescu and Schlud [29] found that voice-activated cobot programming is more efficient than typing and other programming techniques. These papers collectively contribute to the development of human–robot vocal communication systems and highlight the challenges and opportunities in this field.

- *Tactile communications*: The incorporation of tactile feedback mechanisms enhances the haptic dimension of human–cobot collaboration [30]. Tactile interfaces, such as force sensors and haptic feedback devices, enable cobots to perceive and respond to variations in physical interactions [31]. Force sensors can detect unexpected resistance, triggering immediate cessation of movement to prevent collisions or accidents [32]. Haptic feedback devices provide physical sensations to human operators, conveying information about the cobot's state or impending actions [33]. This tactile dimension contributes to a more nuanced and sophisticated collaboration, allowing for a greater degree of trust and coordination between human workers and cobots.

Table 1. Advantages and disadvantages of human–cobot communication methods.

Communication Method	Advantages	Disadvantages
Text Communication	- Provides clarity and precision. - Effective for programming and documentation.	- Slower than verbal interaction. - Higher cognitive load. - Limited adaptability in dynamic tasks.
Visual Communication	- Enhances situational awareness and transparency. - AR prevents errors and offers intuitive guidance. - Provides real-time feedback	- Expensive and complex to implement. - Distracting in high-paced tasks. - Over-reliance may hinder reaction time.
Auditory Communication	- Frees visual and manual attention. - Enables multitasking. - Intuitive for natural interaction.	- Susceptible to noise interference. - Inaccurate speech recognition. - Risk of miscommunication in complex tasks
Tactile Communication	- Improves safety with haptic feedback. - Detects resistance and prevents collisions. - Fosters trust in physical interactions.	- Resource-intensive development. - Limited to physical tasks. - Over-sensitivity may lead to unnecessary operation interruptions.
Multimodal Interaction	- Combines methods for rich, adaptable interactions. - Suited to diverse industrial applications. - Offers natural, versatile communication.	- Prohibitively costly. - Requires substantial computational resources. - Sensory overload can reduce task focus and efficiency.

Human–cobot communication in collaborative assembly systems has a key role in task allocation and safety [34,35]. Human–robot communication can involve various modes,

including verbal communication using speech, multimodal interaction involving head movements, eye gaze, and pointing gestures, and nonverbal cues such as facial expressions and hand gestures [27,36]. Multimodal interaction, which combines different modalities, leads to more natural fixation behavior, improved engagement, and faster reaction to instructions in collaborative tasks [37]. In addition to verbal communication, nonverbal cues could play a very important part in human–robot interaction, allowing for a more natural and inviting experience [38]. The concept of multimodal interaction in human–robot applications acknowledges the synergistic use of different interaction methods, enabling a richer interpretation of human–robot interactions [39]. However, multimodal interaction is not only very costly, but it also takes human attention off the task at hand [39,40]. Javaid et al. [41] review the literature on cobots and describe how they are differentiated from robots. Additionally, they discuss typical cobot features, capabilities, collaboration, and industrial scenarios. Rahman et al. [42] describe the prospects of next-generation cobots in Industry 5.0. Keshvarparast [43] stresses the differences between individual workers and proposes a complex mathematical model to optimize a human–robot collaborative assembly line performance to minimize the cycle time. Cobot task assignment must consider the integration of ergonomics and safety considerations for successful implementation [44].

As to vocal interface, several papers report various pioneering cases. An initial evaluation of robotic vocal commands was performed by [45] using simulation. Programming cobots by voice was reported by [46]. Several papers report vocal communication as part of a multimodal interface [47–50]. An evaluation of the voice interface with Arduino-controlled robots is presented by [51]. A multimodal conversational agent for error-handling is presented in [52] and is mainly dominated by voice communication. Voice-Controlled Pick and Place tasks are described by [53]. Younes et al. [54] showed that synchronizing speech command and gestures can reduce the errors and increase reliability.

Recently, some studies have explored the potential of Natural Language Processing (NLP) for enhancing human–cobot interaction [55,56]. With the advent of generative pretrained technology (GPT), a wave of studies heralded the potential of Large Language Models (LLMs) for improving HCC [57–61]. However, none of them showed real industrial application. Finally, a good tutorial on the use of LLMs for generating a cobot control code is described by [62] and recent surveys on the use of LLMs for interacting with robots are offered by [63–65].

Studies have also dealt with the social and psychological impact of cobots on society and workers. Psychological implications such as anxiety, trust, and the human–cobot relationship remain focal concerns [2,66,67]. A consistent theme across studies is the importance of trust and transparency in cobot decision-making to mitigate stress and resistance [66–68]. The emotional responses of workers to cobots are varied and range from developing attachments (dependency on the cobot) to experiencing competition with the cobots. These responses have brought the need for ethical frameworks for equitable implementation [69]. The perceived competency and usability of cobots also influence acceptance and collaboration quality [66]. Cobot integration is found to impact job satisfaction positively when workers feel safe and involved in decision-making processes [70]. Furthermore, cross-industry studies reveal the importance of cultural considerations in managing automation anxiety and preserving mental well-being [71]. Effective integration strategies, aligned with human-centric policies, are pivotal for maximizing benefits while minimizing disruption.

To conclude, vocal communication methods, such as speech commands, free workers' visual and manual attention, allow multitasking and enhance productivity in dynamic environments. Recent advancements focus on voice programming for cobots, improving the reliability and precision of commands. Synchronized verbal commands with gestures further enhance interaction quality, reducing errors and increasing task efficiency.

Research into vocal enhancements has introduced conversational agents that dominate error handling through speech-based systems, creating a natural and responsive collaborative experience. The integration of verbal interfaces in multimodal systems, combining voice with visual or tactile feedback, offers enriched communication while addressing complex tasks. Ongoing development emphasizes improving speech recognition accuracy, overcoming industrial noise challenges and fostering trust through transparent vocal feedback mechanisms. These advancements ensure verbal and vocal systems remain central to the future of seamless human–cobot interaction.

Recent research highlights NLP and LLMs like GPT for improving human–cobot interaction, though industrial applications are limited. Studies emphasize psychological impacts, including trust, anxiety, and emotional responses like attachment or competition. Trust, usability, and cultural considerations are vital for reducing resistance and managing automation anxiety. Human-centric strategies ensure effective cobot integration and worker well-being.

3. Main Assistive Scenarios

Cobots can assist an industrial worker in many ways and in many types of tasks. Based on the literature, we define the main scenarios that cover large part of these tasks [41] as follows:

- Standard **pick and place** (known locations): There may be several different tasks, and their names and locations should be clearly defined for the human and cobot. The cobot trajectory of a given pick and place is relatively easy to compute. As an input for computing the trajectory, it only needs the coordinates in the space of the origin (pick), the destination (place), and the cobot's end effector (the edge of the cobot's arm). The end effector must be properly quipped for the "grasp" at the pick location and the "release" at the place location.
- **Fetch** distinct **tool/part/material**: Visual search may be needed for identifying the object and corresponding grasping strategy. On the other hand, the destination location must be either predetermined or determined based on the type of tool/part and the context (such as the operation stage or the worker's location). Besides the need to identify the origin and destination coordinates, the trajectory computation carries a striking resemblance to the pick and place task.
- **Return tool/part/material**: A predetermined pickup place typically facilitates finding and picking the tool/part/material. Returning the tool to its dedicated storage location may be easy if there is only one possible location for this tool/part. However, having several identical tools may require a visual search for an empty placing location. Additionally, tool placement may require some manipulation and an aligning strategy.
- **Dispose** of **defective tool/part/material**: The cobot must identify the defective tool/part/material and plan a trajectory for grasping it. Then, it should identify the closest or correct dispose location, and plan a trajectory to reach it.
- **Turn Object**: "Turn object" requires specifying the object, its location, the turn direction, and the turn extent in part of a circle (e.g., half, third. . .tenth of a circle) or degrees. For example, "Turn the left box clockwise three-quarters a of a circle" or "Turn the left box clockwise 270 degrees".
- **Clear away:** This task is required in cases where the cobot is in the path of the main activity or the human worker, and should move away to clear the path. This command must be accompanied by a direction of movement. For example, "Clear away left" turns the whole robot arm left.
- **Move and align**: The input to this task must include a specification of the object to move/align and the destination of the object movement. The end effector camera

should identify the object and the area up to the destination, and plan the trajectory for the move.

- **Drill**: This task must be accompanied by the drill coordinates and orientation, and the specification of drill bit type and drilling speed and depth. The cobot end effector must first be equipped with a drill with the right drill bit. Then, the trajectory to the drilling location is computed. Upon arrival of the drill to its location and orientation a driling operation strats. Finally, the drill returns to its intial position.
- **Screw**: The screw location must be defined (or pre-defined). The "Screw" task assumes that a screw is already in place waiting to be screwed. So, a pick and place of a screw usually precedes this task. To execute this task, first the cobot must locate and fetch a screwdriver. Then, it has to move and align the screwdriver with the screw, and then rotate in the clockwise direction for tightening the screw and counterclockwise for releasing it.
- **Solder/glue-point**: the Location must be pre-defined.
- **Inspect**: The robot has to bring the camera to a given location. Typically, the camera is an integral part of the cobot, located near its end effector. Either the coordinates or the object to be inspected must be specified. In the latter case, dynamic analysis of the camera screenshots has to identify the object and find a good inspection point.
- **Push/Pull**: Care must be taken to clarify the command, as this task may be vague or ambiguous. For example, push/pull the box must be accompanied with a direction and length: "push/pull the box left 10 CM".
- **Press/Detach**: These commands come with either an object or coordinates. In the case of an object (e.g., "press a button" and "detach the handle") the location of the object must be predefined. Otherwise, coordinates must be specified of where to press.
- **Hold** (reach and grasp): Identifying the object for planning the reach, and grasp trajectory.

These core scenarios could benefit from the strategies described in Section 4.

4. Main Strategies for Optimizing Vocal Communication

In this section, we identify various strategies that enhance vocal cobot-related communication.

The relationship between these strategies and the 14 scenarios (from Section 3) are summarized in Table 1. Our proposed strategies are listed as follows:

1. **Workstation map**: Generate a map of the workstation with location-identifying labels of various important points that the cobot arm may need to reach. Store the coordinates of each point in the cobot's control system and hang the map in front of the worker.
2. **Dedicated space for placing tools and parts** (for both the worker and the cobot): Dedicate a convenient place, close to the worker, for the cobot to place or take tools or parts or materials that the worker asked for. This place could serve several tools or parts to be placed from left to right and top to bottom. Store the coordinates of the place in the cobot's control system, and the worker must be informed about this place and understand the expected trajectory of the cobot for reaching this place.
3. **Dedicated storage place** for tool/part: Dedicate a unique storage place for each tool and for part supply that would be easy to reach for both the human worker and the cobot.
4. **Define names for mutual use**: Make sure both the cobot and workers are using the same name for each tool and for each part.
5. **Define predefined trajectories** of the cobot from tool/part areas to the placement area. Thus, the worker knows what movements to expect from the cobot.

These strategies generate a common language for the cobot and the worker, related to locations and objects. They create both clarity and conciseness. Moreover, the trajectory strategy creates worker understanding and anticipation for the cobot moves.

Note that these five strategies appear in Table 2 in abbreviated form as column titles. Each column abbreviated title is explained in the legend under Table 2.

Table 2. Main scenarios and their related proposed strategies.

	Related Strategies (Abbreviated Titles—See Legend Below)				
Main Scenarios	**1 Map**	**2 Placing Space**	**3 Tool/Part Storage**	**4 Defined Names**	**5 Path**
Pick and place	V			V	V
Fetch	V	V	V	V	V
Return	V	V	V	V	V
Dispose	V	V		V	V
Turn predefined				V	V
Turn degree				V	V
Align	V			V	V
Drill	V			V	V
Screw	V		V	V	V
Solder	V			V	V
Inspect		V		V	V
Push/press	V			V	V
Pull/detach	V			V	V
Hold				V	V

Legend for Table 2 abbreviated titles:
Map—workstation map with names for places and objects.
Placing space—dedicated space for placing tools and parts.
Tool/part Storage—dedicated storage place for tools and parts.
Defined Names—defined names for mutual use.
Path—predefined trajectories of the cobot.

5. Illustrative Example

This illustrative example describes a collaborative task in a manufacturing environment where a human worker and a collaborative robot (cobot) work together to assemble electronic components. The workstation and its environment are described in Figure 1. The task involves multiple stages: picking and placing components, fetching necessary tools, returning unused tools, and disposing of defective parts. The aim is to optimize the workflow through efficient vocal communication between the human and the cobot.

Figure 1 was created with the GPT 4.0 and DALL-E2 AI applications and depicts a view of the workspace with storage bins containing various components, the assembly table, the cobot, and the human worker. The cobot is depicted in the process of picking a component from a specific storage bin and placing it on the assembly table where the worker is actively assembling electronic boards.

The assembly task is broken down into four main scenarios: pick and place, fetch, return, and dispose. Each scenario leverages vocal communication strategies to enhance coordination and efficiency. Figure 2 is an enlargement of a part of Figure 1, to clearly designate the locations of “Tool rack B”, “Bin A3” and “Waste bin C” in the shopfloor. Figure 2 was created with the GPT 4.0 and DALL-E2 AI applications.

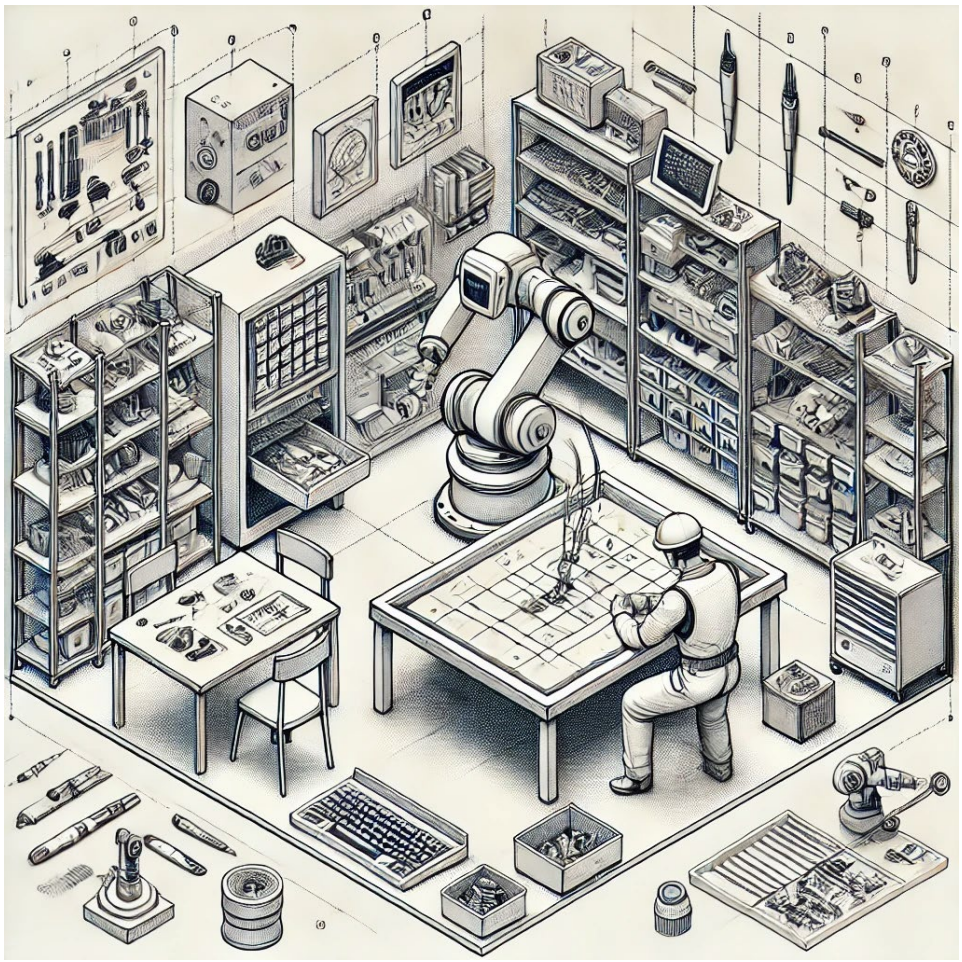


Figure 1. The illustrative example workspace.

5.1. Scenario 1: Pick and Place

The human worker is responsible for assembling electronic boards. The cobot assists by picking components from designated storage bins and placing them on the assembly table. This is carried out using the following three steps.

1. Initial Setup: The workstation map is displayed, showing the locations of various components in storage bins. The cobot's control system stores the coordinates of each bin.
2. Task Execution:
   - Worker: "Cobot, use trajectory "1A" to pick resistor R1 from Bin A3". (Figure 2).
   (a) Cobot: "Picking resistor R1 from Bin A3".
   - Worker: "Cobot, use trajectory "1B" to place it on the spot 1" (on the table)
     - The cobot moves and places it on the designated spot on the assembly table.
   - Cobot: "Resistor R1 placed on the table".
3. Follow-up: The worker continues to assemble the board, instructing the cobot to pick and place other components as needed

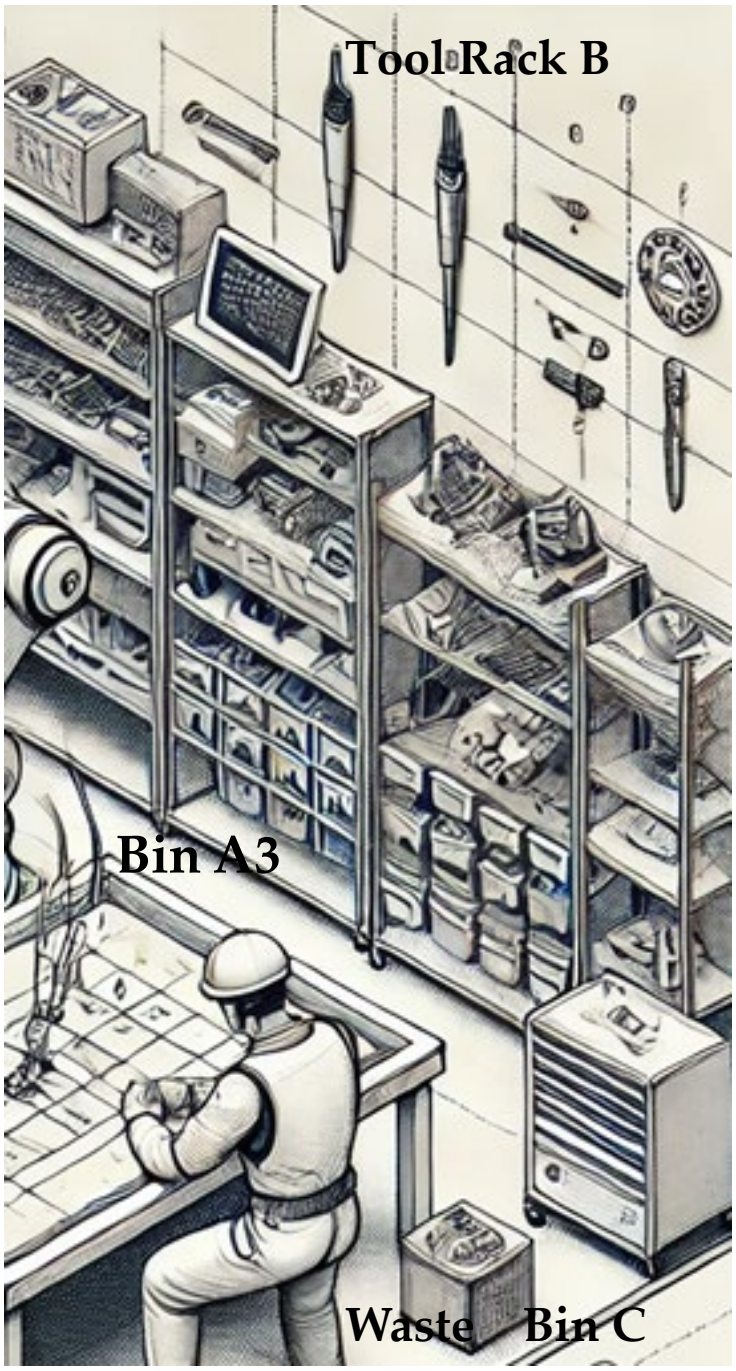


Figure 2. The tool, component, and waste bin locations for the illustrative example.

5.2. Scenario 2: Fetch

During the assembly process, the worker occasionally needs additional tools or components that are not immediately at hand. The cobot assists by fetching these items from storage locations. The system stores in its memory all the standard tool locations for the various storing devices (such as racks, bins, etc.). Thus, the worker does not need to specify the location of the tool and the empty spot on the table. The following steps describe a fetch scenario example.

1. Initial Setup: Dedicated spaces for frequently used tools are identified and labeled on the workstation map. The cobot's control system stores the coordinates for these storage locations.

2. Task Execution:
   - Worker: “Cobot, fetch the soldering iron from Tool Rack B”. (Figure 2). The cobot has the soldering iron coordinates stored in its memory.
   - Cobot: “Fetching soldering iron from Tool Rack B”.
     - The cobot moves to Tool Rack B, identifies the soldering iron, and picks it from Rack B.
     - The cobot identifies an empty space on the table near to the worker and moves to place it on that space.
   - Cobot: “Soldering iron delivered”.
3. Follow-up: The worker uses the soldering iron to solder components onto the board, instructing the cobot to fetch other tools as needed.

5.3. Scenario 3: Return

After using tools or components, the worker may need to return them to their designated storage locations to maintain an organized workspace. The cobot assists by returning these items. This is carried out using the following three steps.

1. Initial Setup:
   - Storage locations for each tool are predefined and stored in the cobot’s control system.
   - The workstation map includes these locations for easy reference.
2. Task Execution:
   - Worker: “Cobot, return the soldering iron to Tool Rack B” (see Figure 2)
   - Cobot: “Returning soldering iron to Tool Rack B”.
   - The cobot takes the soldering iron from the table and returns it to its designated spot in Tool Rack B.
   - Cobot: “Soldering iron returned to Tool Rack B”.
3. Follow-up: The worker continues with the assembly, knowing that the workspace remains organized and tools are readily available for future use.

5.4. Scenario 4: Dispose

Defective components or used materials need to be disposed of properly. The cobot assists by disposing of these items in designated waste bins. This is carried out using the following three steps.

1. Initial Setup:
   - Waste disposal bins are identified and labeled on the workstation map.
   - The cobot’s control system stores the coordinates for these bins.
2. Task Execution:
   - Worker: “Cobot, use trajectory “2” to dispose of this defective capacitor in Waste Bin C”. (Figure 2)
   - Cobot: “Disposing of defective capacitor in Waste Bin C”.
   - The cobot identifies the capacitor and moves its arm above the table, picks up the defective capacitor, and disposes of it in Waste Bin C.
   - Cobot: “Defective capacitor disposed of in Waste Bin C”.
3. Follow-up: The worker continues assembling the electronic board, confident that defective parts are properly disposed of, maintaining a safe and clean workspace.

Throughout the task, the human worker and cobot maintain a continuous, clear line of vocal communication. The cobot’s predefined responses and actions ensure that the worker is always aware of the cobot’s status and actions, fostering a collaborative environment.

This integration not only improves task efficiency but also enhances safety and organization in the workspace. This detailed illustrative example demonstrates how vocal communication strategies can effectively coordinate human and cobot interactions in a manufacturing setting. By optimizing the pick and place, fetch, return, and dispose scenarios, the collaboration between human workers and cobots can significantly enhance productivity, safety, and overall operational efficiency [1–3,11,72–78]. This framework serves as a guide for implementing similar collaborative tasks in various industrial environments.

6. Discussion

Table 1 highlights the crucial role of well-defined communication strategies between human workers and collaborative robots (cobots). Well-defined communication enhances efficiency and safety in collaborative environments. Clear communication protocols, such as defining names for places, objects, actions, and trajectories, are essential for effective bidirectional communication.

A key finding is the importance of cobots conveying predetermined path trajectories to human workers. This ensures workers are aware of cobot movements, which is important for all cobot activities. Creating a comprehensive workstation map detailing locations for both humans and cobots is also vital for successful communication.

Establishing a common language and naming objects and locations fosters a shared understanding of the work environment. This shared understanding optimizes cobot-assisted tasks and allows workers to anticipate cobot movements, enhancing collaborative efficiency.

The strategies in Table 1 contribute to a structured communication framework, ensuring clarity and reducing misunderstandings. These strategies create a sense of order and predictability, further enhancing safety and productivity.

Human workers, unlike cobots, rely on memory for retrieving names, places, objects, maps, and trajectories. Therefore, comprehensive training is imperative before they begin working with cobots. Training should cover technical aspects and the nuances of effective communication and collaboration.

AR applications offer promising tools for communication and collaboration, providing an alternative to traditional workstation maps. AR can simulate collaborative scenarios, preparing workers for real-world interactions with cobots and offering immersive training experiences.

Safety measures are paramount in human–cobot collaboration. Addressing issues such as collision avoidance, force sensing, and emergency stop mechanisms is crucial to prevent accidents and ensure a secure working environment. Implementing safety standards and regulations is essential for governing human–cobot interactions and promoting a safe and productive collaborative workspace.

6.1. Discussion for Scenario 1: Pick and Place

For the pick and place scenario, integrating precise hardware and efficient software is paramount. The following discussion addresses the necessary hardware and software tools for successful implementation.

Hardware:

In terms of hardware, the type of cobot is crucial, as is the type of vision system to be integrated with the cobot. A lightweight yet capable cobot, such as Universal Robots' UR10e or FANUC CRX-10iA, ensures consistent pick-and-place performance. The chosen cobot should be equipped with cameras for 3D spatial recognition such as the Intel RealSense D455 or Ensenso N series.

For pick and place, the end effector is also of very high importance, and also the force sensors. The end effector should be equipped with adaptive grippers (like the

OnRobot RG2 or Robotiq 2F-85) which handle diverse object shapes and weights. Force–torque sensors (like the Robotiq FT300) should be used for accurate force application and collision detection.

Software:

Software components should ensure precision in repetitive tasks, enhancing productivity and worker–cobot collaboration. In particular, task coordination needs a good control platform (e.g., ROS (Robot Operating System)). A vocal interface ensures intuitive communication between cobots and workers (examples of such interfaces are Google Dialogflow or Amazon Lex). For trajectory control and optimization, the MoveIt application is a possible choice. Object recognition could be achieved by TensorFlow or OpenCV for precise object identification and pose estimation.

6.2. Discussion for Scenario 2: Fetch

In the fetch scenario, the cobot retrieves tools or materials efficiently, guided by robust mapping and object identification systems. With the following implementation hardware and software tools, cobots can rapidly locate, fetch, and deliver tools or parts, minimizing human effort.

Hardware:

The fetch scenario is suited for flexible cobots like the ABB YuMi or KUKA LBR iiwa, ideal for environments with spatial constraints. The vision system must be able to identify objects in cluttered storage. Implementation is best in high-resolution 2D or 3D cameras, like the Keyence CV-X or Intel RealSense. Fetch requires a gripper, designed for secure object handling. This is best achieved by multi-grip robotic hands, such as the Schunk Co-act EGP-C gripper.

Software:

Fetch implicitly assumes autonomous navigation and mapping: this could be achieved by simultaneous localization and mapping (SLAM) algorithms for dynamic workstation navigation. Implementing the fetch task may require using Python-based frameworks to integrate workstation maps with cobot actions, automating tool fetch routines. Finally, in case something goes wrong, or if further input is needed, real-time communication becomes important and could be achieved by NLP tools such as Wit.ai or Rasa.

6.3. Discussion for Scenario 3: Return

In the return scenario, the cobot assists the worker by returning items such as tools and parts to their designated locations. Thus, for the return scenario accurate placement and efficient communication are essential. The proposed framework ensures a tidy and efficient workspace while allowing workers to focus on core tasks, improving overall productivity. The implementation discussion on hardware and software is as follows.

Hardware:

For the return scenario, cobots should be capable of handling higher payloads for larger tools. An example of such a cobot is the FANUC CRX-20iA/L. The gripping tool of the end effector has to handle varied tool types and sizes. Robotiq Hand-E grippers offer reliable handling of varied tool types and sizes. Finally, the return scenario would require proximity and touch sensors for safe operations in shared workspaces.

Software:

The return scenario requires a Storage Management system. This system must hold an updated list of all tools, parts, and equipment (items). For each item, the system keeps

its unique ID, type, weight, dimensions, designated storage locations, and the current location. This Storage Management system should be implemented using a relational database-driven storage system that needs to handle frequent real-time updates (of current tool location) and quick queries mostly related to the item's location or its predefined location/s. SQL-based systems provide good alternative solutions. For example, MySQL, PostgreSQL, or Microsoft SQL Server are all viable solutions. For the return scenario, path planning is also required: defining and executing optimal return paths may use MoveIt or the MATLAB Robotics Toolbox. Finally, in case something goes wrong, or if further input is needed, real-time communication becomes important and could be achieved by interactive vocal guidance for rapid and clear instructions using a voice recognition system such as Azure Speech Service.

6.4. Discussion on Social and Psychological Impact of Cobots on Society and Workers

Based on references [66–71], AI and cobot integration introduces challenges such as varying emotional responses, with some workers forming positive relationships and others experiencing anxiety about job displacement. Cobots significantly influence workplace dynamics, reshaping team interactions and requiring thoughtful implementation to improve job satisfaction and engagement. Organizational and cultural factors play a pivotal role in shaping workers' acceptance, with ethical considerations like fairness and transparency being crucial to building trust. Human-centric designs and strategies that prioritize psychological well-being and adaptability are essential for successful cobot adoption, ensuring they complement rather than compete with human labor.

In conclusion, effective communication and collaboration are critical in human–cobot interactions. The identified strategies provide practical insights for optimizing vocal communication and enhancing productivity and safety in cobot-assisted tasks. Continuous research and development are vital for staying abreast of evolving technologies and refining communication strategies. This manuscript contributes to this dynamic field by offering comprehensive insights and strategies, guiding future research and industry practices in collaborative robotics.

7. Conclusions

This study highlights the important role of vocal communication in increasing productivity and safety in cobot-assisted tasks. This paper emphasizes the importance of common language for describing places, objects, and trajectories to enhance task efficiency and foster a secure working environment. Acknowledging the significance of comprehensive worker training and the potential of AR applications, this research underscores the need for a holistic approach to human–cobot collaboration. In particular, some of the useful lessons learned from this study are as follows:

- Most cobot tasks could be classified into a small group of task types, where each type has characteristic movements (14 main types identified by [41]).
- A three-dimensional map of the workstation that is understood by a human and enables the translation of places and trajectories to 3D coordinates is a key enabler of human–robot collaboration.
- Common expressions between the human worker and the digital system are essential for collaboration. These expressions could be task type, points in the workstation, pre-agreed-upon trajectories, gripping instructions, etc.
- Setting dedicated places for tools and intermediate storage facilitates human–cobot collaboration.

This manuscript contributes to the evolving field of human–robot collaboration by providing a comprehensive exploration of strategies to enhance productivity and safety

in assembly environments. By synthesizing theoretical frameworks, practical recommendations, and empirical evidence, this work aims to guide future research and industry practices in the dynamic field of collaborative robotics.

Future research could explore ways to refine communication protocols and explore virtual reality (VR) and AR for better worker training and improved safety. Another direction may examine the integration of LLMs into human–robotic communications. Investigating the impact of the suggested strategies (generating a workstation map; defining a placement space; defining tool/part storage; defining places and objects names; and defining trajectories and paths) in a variety of industrial settings and learning to improve collaboration will contribute to the dynamic field of collaborative robotics. As technology advances, understanding the dynamics of human–robot interactions remains crucial for unlocking the full potential of cobots in Industry 5.0. This study serves as a foundation, urging researchers and practitioners to explore further and implement innovative solutions for the seamless integration of cobots into modern manufacturing.

Author Contributions: Conceptualization, Y.C. and M.F.; Methodology, Y.C. and S.R.; Investigation, M.F.; Formal analysis, Y.C.; Validation, S.R.; Writing—original draft preparation, Y.C. and S.R.; Writing—review and editing, Y.C. and M.F.; Visualization, S.R. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: Not applicable.

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Faccio, M.; Cohen, Y. Intelligent Cobot Systems: Human-Cobot Collaboration in Manufacturing. *J. Intell. Manuf.* **2023**, *35*, 1905–1907. [CrossRef]
2. Faccio, M.; Granata, I.; Menini, A.; Milanese, M.; Rossato, C.; Bottin, M.; Minto, R.; Pluchino, P.; Gamberini, L.; Boschetti, G. Human Factors in Cobot Era: A Review of Modern Production Systems Features. *J. Intell. Manuf.* **2023**, *34*, 85–106. [CrossRef]
3. Liu, L.; Schoen, A.J.; Henrichs, C.; Li, J.; Mutlu, B.; Zhang, Y.; Radwin, R.G. Human Robot Collaboration for Enhancing Work Activities. *Hum. Factors* **2024**, *66*, 158–179. [CrossRef] [PubMed]
4. Papetti, A.; Ciccarelli, M.; Scoccia, C.; Palmieri, G.; Germani, M. A Human-Oriented Design Process for Collaborative Robotics. *Int. J. Comput. Integr. Manuf.* **2023**, *36*, 1760–1782. [CrossRef]
5. Gross, S.; Krenn, B. A Communicative Perspective on Human–Robot Collaboration in Industry: Mapping Communicative Modes on Collaborative Scenarios. *Int. J. Soc. Robot.* **2023**, *16*, 1–18. [CrossRef]
6. Moore, B.A.; Urakami, J. The Impact of the Physical and Social Embodiment of Voice User Interfaces on User Distraction. *Int. J. Hum. Comput. Stud.* **2022**, *161*, 102784. [CrossRef]
7. Marklin Jr, R.W.; Toll, A.M.; Bauman, E.H.; Simmins, J.J.; LaDisa, J.F., Jr.; Cooper, R. Do Head-Mounted Augmented Reality Devices Affect Muscle Activity and Eye Strain of Utility Workers Who Do Procedural Work? Studies of Operators and Manhole Workers. *Hum. Factors* **2022**, *64*, 305–323. [CrossRef]
8. Heydaryan, S.; Suaza Bedolla, J.; Belingardi, G. Safety Design and Development of a Human-Robot Collaboration Assembly Process in the Automotive Industry. *Appl. Sci.* **2018**, *8*, 344. [CrossRef]
9. Petzoldt, C.; Harms, M.; Freitag, M. Review of Task Allocation for Human-Robot Collaboration in Assembly. *Int. J. Comput. Integr. Manuf.* **2023**, *36*, 1675–1715. [CrossRef]
10. Schmidbauer, C.; Zafari, S.; Hader, B.; Schlund, S. An Empirical Study on Workers' Preferences in Human–Robot Task Assignment in Industrial Assembly Systems. *IEEE Trans. Hum. Mach. Syst.* **2023**, *53*, 293–302. [CrossRef]
11. Jacomini Prioli, J.P.; Liu, S.; Shen, Y.; Huynh, V.T.; Rickli, J.L.; Yang, H.-J.; Kim, S.-H.; Kim, K.-Y. Empirical Study for Human Engagement in Collaborative Robot Programming. *J. Integr. Des. Process Sci.* **2023**, *26*, 159–181. [CrossRef]

12. Scalise, R.; Rosenthal, S.; Srinivasa, S. Natural Language Explanations in Human-Collaborative Systems. In Proceedings of the Companion of the 2017 ACM/IEEE International Conference on Human-Robot Interaction, Vienna, Austria, 6–9 March 2017; pp. 377–378.
13. Kontogiorgos, D. Utilising Explanations to Mitigate Robot Conversational Failures. *arXiv* **2023**, arXiv:2307.04462.
14. Zieliński, K.; Walas, K.; Heredia, J.; Kjærgaard, M.B. A Study of Cobot Practitioners Needs for Augmented Reality Interfaces in the Context of Current Technologies. In Proceedings of the 2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN), Vancouver, Canada, 8–12 August 2021; pp. 292–298.
15. Pascher, M.; Kronhardt, K.; Franzen, T.; Gruenefeld, U.; Schneegass, S.; Gerken, J. My Caregiver the Cobot: Comparing Visualization Techniques to Effectively Communicate Cobot Perception to People with Physical Impairments. *Sensors* **2022**, *22*, 755. [CrossRef] [PubMed]
16. Eimontaite, I.; Cameron, D.; Rolph, J.; Mokaram, S.; Aitken, J.M.; Gwilt, I.; Law, J. Dynamic Graphical Instructions Result in Improved Attitudes and Decreased Task Completion Time in Human–Robot Co-Working: An Experimental Manufacturing Study. *Sustainability* **2022**, *14*, 3289. [CrossRef]
17. Carriero, G.; Calzone, N.; Sileo, M.; Pierri, F.; Caccavale, F.; Mozzillo, R. Human-Robot Collaboration: An Augmented Reality Toolkit for Bi-Directional Interaction. *Appl. Sci.* **2023**, *13*, 11295. [CrossRef]
18. Sauer, V.; Sauer, A.; Mertens, A. Zoomorphic Gestures for Communicating Cobot States. *IEEE Robot. Autom. Lett.* **2021**, *6*, 2179–2185. [CrossRef]
19. Deuerlein, C.; Langer, M.; Seßner, J.; Heß, P.; Franke, J. Human-robot-interaction using cloud-based speech recognition systems. *Procedia Cirp* **2021**, *97*, 130–135. [CrossRef]
20. Turri, S.; Rizvi, M.; Rabini, G.; Melonio, A.; Gennari, R.; Pavani, F. Orienting Auditory Attention through Vision: The Impact of Monaural Listening. *Multisensory Res.* **2021**, *35*, 1–28. [CrossRef]
21. Resing, M. Industrial Cobot Sound Design Study: Audio Design Principles for Industrial Human-Robot Interaction. Master's Thesis, University of Twente, Enschede, The Netherlands, 2023.
22. Tran, N. *Exploring Mixed Reality Robot Communication under Different Types of Mental Workload*; Colorado School of Mines: Golden, CO, USA, 2020; ISBN 9798645495275.
23. Telkes, P.; Angleraud, A.; Pieters, R. Instructing Hierarchical Tasks to Robots by Verbal Commands. In Proceedings of the 2024 IEEE/SICE International Symposium on System Integration (SII), Ha Long, Vietnam, 8–11 January 2024; pp. 1139–1145.
24. Salehzadeh, R.; Gong, J.; Jalili, N. Purposeful Communication in Human–Robot Collaboration: A Review of Modern Approaches in Manufacturing. *IEEE Access* **2022**, *10*, 129344–129361. [CrossRef]
25. Rusan, H.-A.; Mocanu, B. Human-Computer Interaction Through Voice Commands Recognition. In Proceedings of the 2022 International Symposium on Electronics and Telecommunications, Timisoara, Romania, 10–11 November 2022; pp. 1–4.
26. Carr, C.; Wang, P.; Wang, S. A Human-Friendly Verbal Communication Platform for Multi-Robot Systems: Design and Principles. In *UK Workshop on Computational Intelligence*; Springer: Berlin/Heidelberg, Germany, 2023; pp. 580–594.
27. McMillan, D.; Jaber, R.; Cowan, B.R.; Fischer, J.E.; Irfan, B.; Cumbal, R.; Zargham, N.; Lee, M. Human-Robot Conversational Interaction (HRCI). In Proceedings of the Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction, Stockholm, Sweden, 13–16 March 2023; pp. 923–925.
28. Lee, K.M.; Krishna, A.; Zaidi, Z.; Paleja, R.; Chen, L.; Hedlund-Botti, E.; Schrum, M.; Gombolay, M. The Effect of Robot Skill Level and Communication in Rapid, Proximate Human-Robot Collaboration. In Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction, Stockholm, Sweden, 13–16 March 2023; pp. 261–270.
29. Ionescu, T.B.; Schlund, S. Programming Cobots by Voice: A Human-Centered, Web-Based Approach. *Procedia CIRP* **2021**, *97*, 123–129. [CrossRef]
30. Sorgini, F.; Farulla, G.A.; Lukic, N.; Danilov, I.; Roveda, L.; Milivojevic, M.; Pulikottil, T.B.; Carrozza, M.C.; Prinetto, P.; Tolio, T. Tactile Sensing with Gesture-Controlled Collaborative Robot. In Proceedings of the 2020 IEEE International Workshop on Metrology for Industry 4.0 & IoT, Rome, Italy, 3–5 June 2020; pp. 364–368.
31. Guda, V.; Mugisha, S.; Chevallereau, C.; Zoppi, M.; Molfino, R.; Chablat, D. Motion Strategies for a Cobot in a Context of Intermittent Haptic Interface. *J. Mech. Robot.* **2022**, *14*, 041012. [CrossRef]
32. Zurlo, D.; Heitmann, T.; Morlock, M.; De Luca, A. Collision Detection and Contact Point Estimation Using Virtual Joint Torque Sensing Applied to a Cobot. In Proceedings of the 2023 IEEE International Conference on Robotics and Automation (ICRA), London, UK, 29 May–2 June 2023; 2023; pp. 7533–7539.
33. Costes, A.; Lécuyer, A. Inducing Self-Motion Sensations with Haptic Feedback: State-of-the-Art and Perspectives on "Haptic Motion". *IEEE Trans. Haptics* **2023**, *16*, 171–181. [CrossRef] [PubMed]
34. Keshvarparast, A.; Battini, D.; Battaia, O.; Pirayesh, A. Collaborative Robots in Manufacturing and Assembly Systems: Literature Review and Future Research Agenda. *J. Intell. Manuf.* **2023**, *35*, 2065–2118. [CrossRef]
35. Calzavara, M.; Faccio, M.; Granata, I.; Trevisani, A. Achieving productivity and operator well-being: A dynamic task allocation strategy for collaborative assembly systems in Industry 5.0. *Int. J. Adv. Manuf. Technol.* **2024**, *134*, 3201–3216. [CrossRef]

36. Schreiter, T.; Morillo-Mendez, L.; Chadalavada, R.T.; Rudenko, A.; Billing, E.; Magnusson, M.; Arras, K.O.; Lilienthal, A.J. Advantages of Multimodal versus Verbal-Only Robot-to-Human Communication with an Anthropomorphic Robotic Mock Driver. In Proceedings of the 2023 32nd IEEE International Conference on Robot and Human Interactive Communication (RO-MAN), Busan, South Korea, 28–31 August 2023; pp. 293–300.
37. Rautiainen, S.; Pantano, M.; Traganos, K.; Ahmadi, S.; Saenz, J.; Mohammed, W.M.; Martinez Lastra, J.L. Multimodal Interface for Human–Robot Collaboration. *Machines* **2022**, *10*, 957. [CrossRef]
38. Urakami, J.; Seaborn, K. Nonverbal Cues in Human–Robot Interaction: A Communication Studies Perspective. *ACM Trans. Hum. Robot Interact.* **2023**, *12*, 1–21. [CrossRef]
39. Park, K.-B.; Choi, S.H.; Lee, J.Y.; Ghasemi, Y.; Mohammed, M.; Jeong, H. Hands-Free Human–Robot Interaction Using Multimodal Gestures and Deep Learning in Wearable Mixed Reality. *IEEE Access* **2021**, *9*, 55448–55464. [CrossRef]
40. Nagrani, A.; Yang, S.; Arnab, A.; Jansen, A.; Schmid, C.; Sun, C. Attention Bottlenecks for Multimodal Fusion. *Adv. Neural Inf. Process. Syst.* **2021**, *34*, 14200–14213.
41. Javaid, M.; Haleem, A.; Singh, R.P.; Rab, S.; Suman, R. Significant Applications of Cobots in the Field of Manufacturing. *Cogn. Robot.* **2022**, *2*, 222–233. [CrossRef]
42. Rahman, M.M.; Khatun, F.; Jahan, I.; Devnath, R.; Bhuiyan MA, A. Cobotics: The Evolving Roles and Prospects of Next-Generation Collaborative Robots in Industry 5.0. *J. Robot.* **2024**, *2024*, 2918089. [CrossRef]
43. Keshvarparast, A.; Katiraee, N.; Finco, S.; Calzavara, M. Integrating collaboration scenarios and workforce individualization in collaborative assembly line balancing. *Int. J. Prod. Econ.* **2024**, *279*, 109450. [CrossRef]
44. Ragil, S.M.; Tieling, Z.; Kiridena, S. Overview of Ergonomics and Safety Aspects of Human-Cobot Interaction in the Manufacturing Industry. *Int. Conf. Inform. Technol. Eng.* **2023**, *21*, 401–405.
45. Adamini, R.; Antonini, N.; Borboni, A.; Medici, S.; Nuzzi, C.; Pagani, R.; Tonola, C. User-friendly human-robot interaction based on voice commands and visual systems. In Proceedings of the 2021 24th International Conference on Mechatronics Technology (ICMT), Singapore, 18–22 December 2021; p. 1.
46. Ionescu, T.B.; Schlund, S. Programming cobots by voice: A pragmatic, web-based approach. *Int. J. Comput. Integr. Manuf.* **2023**, *36*, 86–109. [CrossRef]
47. Rouillard, J.; Vannobel, J.M. Multimodal Interaction for Cobot Using MQTT. *Multimodal Technol. Interact.* **2023**, *7*, 78. [CrossRef]
48. Su, H.; Qi, W.; Chen, J.; Yang, C.; Sandoval, J.; Laribi, M.A. Recent advancements in multimodal human–robot interaction. *Front. Neurorobotics* **2023**, *17*, 1084000. [CrossRef]
49. Ferrari, D.; Pupa, A.; Signoretti, A.; Secchi, C. Safe Multimodal Communication in Human-Robot Collaboration. In *In International Workshop on Human-Friendly Robotics*; Springer Nature: Cham, Switzerland, 2023; pp. 151–163.
50. D'Attanasio, S.; Alabert, T.; Francis, C.; Studzinska, A. Exploring Multimodal Interactions with a Robot Assistant in an Assembly Task: A Human-Centered Design Approach. In Proceedings of the 8th International Conference on Human Computer Interaction Theory and Applications, Rome, Italy, 27–29 February 2024; pp. 549–556.
51. Abdulrazzaq, A.Z.; Ali, Z.G.; Al-Ani AR, M.; Khaleel, B.M.; Alsalame, S.; Snovyda, V.; Kanbar, A.B. Evaluation of Voice Interface Integration with Arduino Robots in 5G Network Frameworks. In Proceedings of the 2024 36th Conference of Open Innovations Association (FRUCT), Lappeenranta, Finland, 24–26 April 2024; pp. 44–55. [CrossRef]
52. Schwarz, D.; Zarcone, A.; Laquai, F. Talk to your Cobot: Faster and more efficient error-handling in a robotic system with a multi-modal Conversational Agent. *Proc. Mensch und Comput.* **2024**, *2024*, 520–532.
53. Asha, C.S.; D'Souza, J.M. Voice-Controlled Object Pick and Place for Collaborative Robots Employing the ROS2 Framework. In Proceedings of the 2024 International Conference on Advances in Modern Age Technologies for Health and Engineering Science (AMATHE), Shivamogga, India, 16–17 May 2024; pp. 1–7.
54. Younes, R.; Elisei, F.; Pellier, D.; Bailly, G. Impact of verbal instructions and deictic gestures of a cobot on the performance of human coworkers. In Proceedings of the IEEE-RAS International Conference on Humanoid Robots, Nancy, France, 22–24 November 2024.
55. Siwach, G.; Li, C. Enhancing Human Cobot Interaction using Natural Language Processing. In Proceedings of the 2023 IEEE 4th International Multidisciplinary Conference on Engineering Technology (IMCET), Beirut, Lebanon, 12–14 December 2023; pp. 21–26.
56. Siwach, G.; Li, C. Unveiling the Potential of Natural Language Processing in Collaborative Robots (Cobots): A Comprehensive Survey. In Proceedings of the 2024 IEEE International Conference on Consumer Electronics (ICCE), Las Vegas, NV, USA, 6–8 January 2024; pp. 1–6.
57. Gkournelos, C.; Konstantinou, C.; Angelakis, P.; Michalos, G.; Makris, S. Enabling Seamless Human-Robot Collaboration in Manufacturing Using LLMs. In *European Symposium on Artificial Intelligence in Manufacturing*; Springer Nature: Cham, Switzerland, 2023; pp. 81–89.
58. Lakhnati, Y.; Pascher, M.; Gerken, J. Exploring a GPT-based large language model for variable autonomy in a VR-based human-robot teaming simulation. *Front. Robot. AI* **2024**, *11*, 1347538. [CrossRef]

59. Kim, C.Y.; Lee, C.P.; Mutlu, B. Understanding large-language model (llm)-powered human-robot interaction. In Proceedings of the 2024 ACM/IEEE International Conference on Human-Robot Interaction, Boulder, CO, USA, 11–15 March 2024; pp. 371–380.
60. Ye, Y.; You, H.; Du, J. Improved trust in human-robot collaboration with ChatGPT. *IEEE Access* **2023**, *11*, 55748–55754. [CrossRef]
61. Wang, T.; Fan, J.; Zheng, P. An LLM-based vision and language cobot navigation approach for Human-centric Smart Manufacturing. *J. Manuf. Syst.* **2024**, *75*, 299–305. [CrossRef]
62. Kim, Y.; Kim, D.; Choi, J.; Park, J.; Oh, N.; Park, D. A survey on integration of large language models with intelligent robots. *Intell. Serv. Robot.* **2024**, *17*, 1091–1107. [CrossRef]
63. Zhang, C.; Chen, J.; Li, J.; Peng, Y.; Mao, Z. Large language models for human-robot interaction: A review. *Biomim. Intell. Robot.* **2023**, *3*, 100131. [CrossRef]
64. Jeong, H.; Lee, H.; Kim, C.; Shin, S. A Survey of Robot Intelligence with Large Language Models. *Appl. Sci.* **2024**, *14*, 8868. [CrossRef]
65. Shi, Z.; Landrum, E.; O'Connell, A.; Kian, M.; Pinto-Alva, L.; Shrestha, K.; Zhu, X.; Matarić, M.J. How Can Large Language Models Enable Better Socially Assistive Human-Robot Interaction: A Brief Survey. *Proc. AAAI Symp. Ser.* **2024**, *3*, 401–404. [CrossRef]
66. Kawaharazuka, K.; Matsushima, T.; Gambardella, A.; Guo, J.; Paxton, C.; Zeng, A. Real-world robot applications of foundation models: A review. *Adv. Robot.* **2024**, *38*, 1232–1254. [CrossRef]
67. Liao, S.; Lin, L.; Chen, Q. Research on the acceptance of collaborative robots for the industry 5.0 era—The mediating effect of perceived competence and the moderating effect of robot use self-efficacy. *Int. J. Ind. Ergon.* **2023**, *95*, 103455. [CrossRef]
68. Nenna, F.; Zanardi, D.; Maria Orlando, E.; Mingardi, M.; Buodo, G.; Gamberini, L. Addressing Trust and Negative Attitudes Toward Robots in Human-Robot Collaborative Scenarios: Insights from the Industrial Work Setting. In Proceedings of the 17th International Conference on PErvasive Technologies Related to Assistive Environments, Crete, Greece, 26–28 June 2024; pp. 259–264.
69. Pietrantoni, L.; Favilla, M.; Fraboni, F.; Mazzoni, E.; Morandini, S.; Benvenuti, M.; De Angelis, M. Integrating collaborative robots in manufacturing, logistics, and agriculture: Expert perspectives on technical, safety, and human factors. *Front. Robot. AI* **2024**, *11*, 1342130. [CrossRef]
70. Weiss, A.; Wortmeier, A.-K.; Kubicek, B. Cobots in Industry 4.0: A Roadmap for Future Practice Studies on Human–Robot Collaboration. *IEEE Trans. Hum. Mach. Syst.* **2021**, *51*, 335–345. [CrossRef]
71. Paliga, M. The relationships of human-cobot interaction fluency with job performance and job satisfaction among Cobot operators—The moderating role of workload. *Int. J. Environ. Res. Public Health* **2023**, *20*, 5111. [CrossRef]
72. Fraboni, F.; Brendel, H.; Pietrantoni, L. Evaluating organizational guidelines for enhancing psychological well-being, safety, and performance in technology integration. *Sustainability* **2023**, *15*, 8113. [CrossRef]
73. Liu, L.; Guo, F.; Zou, Z.; Duffy, V.G. Application, development and future opportunities of collaborative robots (cobots) in manufacturing: A literature review. *Int. J. Hum. Comput. Interact.* **2024**, *40*, 915–932. [CrossRef]
74. Yenjai, N.; Dancholvichit, N. Optimizing pick-place operations: Leveraging k-means for visual object localization and decision-making in collaborative robots. *J. Appl. Res. Sci. Tech (JARST)* **2024**, *23*, 254153. [CrossRef]
75. Cohen, Y.; Shoval, S.; Faccio, M.; Minto, R. Deploying cobots in collaborative systems: Major considerations and productivity analysis. *Int. J. Prod. Res.* **2022**, *60*, 1815–1831. [CrossRef]
76. De Simone, V.; Di Pasquale, V.; Giubileo, V.; Miranda, S. Human-Robot Collaboration: An analysis of worker's performance. *Procedia Comput. Sci.* **2022**, *200*, 1540–1549. [CrossRef]
77. Bi, Z.M.; Luo, C.; Miao, Z.; Zhang, B.; Zhang, W.J.; Wang, L. Safety assurance mechanisms of collaborative robotic systems in manufacturing. *Robot. Comput. Integr. Manuf.* **2021**, *67*, 102022. [CrossRef]
78. Vemuri, N.; Thaneeru, N. Enhancing Human-Robot Collaboration in Industry 4.0 with AI-driven HRI. *Power Syst. Tech.* **2023**, *47*, 341–358. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.